Non-Anticipative Market Field Calculus: An Auditable Multiscale Representation for Option-Path Decision Support

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Abstract

We introduce Market Field, a non-anticipative multiscale representation of market path organization for inspectable research and prospective option-decision instrumentation. Completed OHLCV prefixes become a bounded horizon–time pressure surface, time/log-horizon measurements, a 15-coordinate state vector, and a deterministic request-local Form dictionary with separated fit, calibration, and evaluation segments. Across 32 daily prefix endpoints (6,688 related serialized scalars) and a nine-timeframe supplement (46 endpoints; 24,472 full-precision values), tested live channels have zero observed prefix mismatch; all eight fixed-input dictionaries replay exactly. Controlled paths produce ordered horizon-response delays (ρ = 1.000). A new five-coordinate, single-horizon EMA/path/variation/volume baseline uses the same chronology and clustering gate: it accepts multi-Form codebooks for 8/8 assets with median fit silhouette 0.448, versus 2/8 and 0.000 for Market Field. Thus the current evidence does not show superior unsupervised separation. On long-history SPY, entropy windows 8–96 preserve two Forms and yield assignment adjusted Rand indexes 0.878–0.908 against window 24, but the upper calibration-distance-tail rate still varies from 10.6% to 16.7%. The supported contribution is therefore a reconstructible systems-and-methods pipeline, negative and sensitivity evidence, and a zero-direct-field/no-execution path for prospective study—not learned latent states, prediction, or option profitability. Historical field cohorts may participate indirectly inside a separately governed outcome-learning canary whose total weight is capped at 10%; this is instrumentation, not efficacy evidence. Source, derived artifacts, executed notebooks, and hashes are provided; provider restrictions prevent redistribution of raw snapshots.

1 Introduction

Market analysis interfaces usually compress a path into a small set of single-window indicators. This makes the output easy to scan but obscures whether a move is coherent across horizons, confined to one scale, accelerating, fragmenting, or changing its relationship to activity and liquidity. The opposite response—adding more indicators—often produces a wall of numbers with no stable visual or semantic grammar. Both patterns become especially problematic for options: a volatility-dislocation thesis and the underlying price path are distinct objects, yet they are often blended into one opaque score.

We study a narrower systems question: can a hand-engineered multiscale state representation remain non-anticipative, reconstructible, visually legible, and safe to instrument prospectively before economic efficacy is known? Our answer is Market Field, an engineered field over time t and lookback horizon h. It is not learned end-to-end. The word field describes the data structure; pressure and its derivatives are bounded discrete measurements rather than literal forces or hidden physical dimensions.

The system evolved from SwamiChart-style horizon strips (Ehlers & Way, 2012a;b) through a signed pressure surface, a derivative and scale geometry, a human translation layer, and finally a versioned shadow integration for option research. This progression preserves the original multiscale visual intuition while replacing indicator voting with an explicit measurement pipeline.

Version boundary. market_field_calculus_v1 fixes the formulas. Additive semantic_revision=1.3 adds finite-computation/dependency-support coverage, computation identities, and direct/indirect authority receipts while carrying forward semantic 1.2 alignment; legacy payloads remain readable. maturity is only a compatibility alias, not convergence. Relative Field Pair v1 is a separate two-instrument comparison receipt over independently computed formula-v1 fields; it does not change either component field. market_field_preliminary_v2 names the evaluation harness, not a v2 field model.

What this paper claims. The equations and chronology are reconstructible; selected live outputs are prefix-measurable at every tested endpoint; fixed inputs replay deterministically; controlled paths produce declared construction behavior; and direct field snapshots can be logged prospectively with zero scanner weight and no execution authority. What it does not claim. The Forms are not natural latent states, the 15-coordinate representation is not shown to outperform a five-coordinate baseline, and no result establishes prediction, profitability, causal market structure, or autonomous decision quality.

Name	What changes	What does not change
Formula v1	Field and 15-coordinate vector	Fixed within this paper
Semantic 1.3	Coverage, identities, authority receipts	Formula values and stored history
Preliminary v2	Evaluation harness	Not a v2 field model

Table 1: Version names used in the paper.

Our contributions are:

1. a fully specified bounded surface and 15-coordinate multiscale state vector over causal time and log-horizon coordinates;
2. a deterministic translation dictionary with separated proper fit, held-out distance calibration, and evaluation chronology;
3. a reproducible diagnostic suite that includes full-precision prefix tests, controlled paths, a matched cheap baseline, and downstream entropy sensitivity; and
4. a prospective option-research interface that stores immutable point-in-time snapshots under zero-direct-field and no-execution safeguards, with bounded historical-learning attribution.

2 Relationship to Prior Work

Multiscale representation. SwamiCharts arrange an indicator across many lookbacks (Ehlers & Way, 2012a;b). Wavelets, scattering transforms, and multifractal models offer principled multiscale representations (Mallat, 2012; Andén & Mallat, 2014; Bacry et al., 2001). Market Field shares their multiscale motivation but does not implement a wavelet transform, scattering network, or multifractal estimator. Its horizon coordinate is a set of EMA/true-range/path-efficiency measurements selected for causal interactive computation.

State and phase descriptions. Markov switching, structural-break, and online change-point models estimate latent or discontinuous states (Hamilton, 1989; Bai & Perron, 1998; Adams & MacKay, 2007). Market Field instead creates a calibration-relative codebook with deterministic clustering. Its scope plots are engineered two-dimensional projections. They are visually related to phase-space and recurrence analysis (Takens, 1981; Eckmann et al., 1987; Marwan et al., 2007), but they are not delay-coordinate reconstructions, recurrence plots, attractor tests, or cycle detectors. Permutation entropy is the one directly adopted nonlinear statistic (Bandt & Pompe, 2002).

Technical and option context. Prior work gives statistical definitions to technical patterns and price barriers (Lo et al., 2000; Chung & Bellotti, 2021). The present support and resistance layer is deliberately simpler: shifted 20-bar extrema used as causal context, not learned order-book structure. Implied-versus-realized volatility and variance-risk-premium measures are established (Carr & Wu, 2009; Bollerslev et al., 2009; Goyal & Saretto, 2009). Our open question is whether separately measured path fit conditions such optionality signals, not whether the volatility spread itself is novel. Similarly, cross-market dependence and connectedness are well studied (Diebold & Yilmaz, 2012; Baruník & Křehlík, 2018); the current cached-source atlas is an association screen, not causal discovery.

Validation discipline. Financial feature search is exposed to dependence, selection, and multiple testing (White, 2000; Harvey et al., 2016; Bailey et al., 2017). We therefore separate representation diagnostics from economic evaluation. The repository now contains a frozen development-only rolling-origin harness with purging, dependence-aware resampling, multiplicity correction, naive, technical, fixed-break, and two-state HMM comparators, and family ablations; a no-touch prospective holdout and cost-aware option evaluation remain future work. This distinction is central: a correctly non-anticipative implementation can still have no predictive or economic value.

3 Non-Anticipative Market Field

3.1 Pressure over time and horizon

Let O_t, H_t, L_t, C_t, V_t denote a sorted, deduplicated OHLCV prefix, let $M_t = (H_t + L_t)/2$, and index an ordered horizon grid by $h_1 < \cdots < h_J$. For row j, a gap-aware true range normalizes the displacement between a fast EMA with span $\max(2, \lfloor h_j/2 \rfloor)$ and an EMA with span h_j:

$$z_{j,t} = \frac{\mathrm{EMA}_{\max(2,\lfloor h_j/2\rfloor)}(M)_t - \mathrm{EMA}_{h_j}(M)_t}{\mathrm{Mean}_{h_j}(TR)_t}, \qquad d_{j,t} = \frac{z_{j,t}}{1+|z_{j,t}|}, \tag{1}$$

$$e_{j,t} = \mathrm{clip}_{[0,1]} \frac{|M_t - M_{t-h_j}|}{\sum_{i=t-h_j+1}^{t} |M_i - M_{i-1}|}. \tag{2}$$

Equation 2 is Kaufman's efficiency-ratio construction (Kaufman, 1995). The raw signed state $d_{j,t}e_{j,t}$ is smoothed causally in time and blended only with adjacent rows to obtain $P_{j,t}$. Thus P is bounded, directional, and attenuated when net displacement is small relative to path length. Row adjacency is part of v1: it is not a log-distance kernel, so changing horizon density changes the effective model as well as its discretization. The blend is analytical, and all downstream measurements are calculated from the blended surface. Exact coefficients, initialization, zero conventions, and edge behavior appear in Appendix A.

Remark 1 (boundedness and prefix measurability). Let $\mathcal{H} = (h_1, \ldots, h_J)$ be fixed. Assume finite valid OHLC inputs, positive closes, the numerical conventions in Appendix A, and EWM recursions initialized from the observed prefix only. Then $e_{j,t} \in [0,1]$, $d_{j,t}, P_{j,t}, D_q(X)_{j,t} \in (-1,1)$ except for their defined zero cases, clipped unsigned channels lie in $[0,1]$, and every live field output at t is a deterministic function of the OHLCV prefix $X_{1:t}$.

The argument is immediate from convex bounded transforms and prefix recursions; Appendix A.1 records it for completeness. The statement covers the live field and point-in-time option snapshot, not the window-refit dictionary, retrospective atlas, cross-market cache, or future labels.

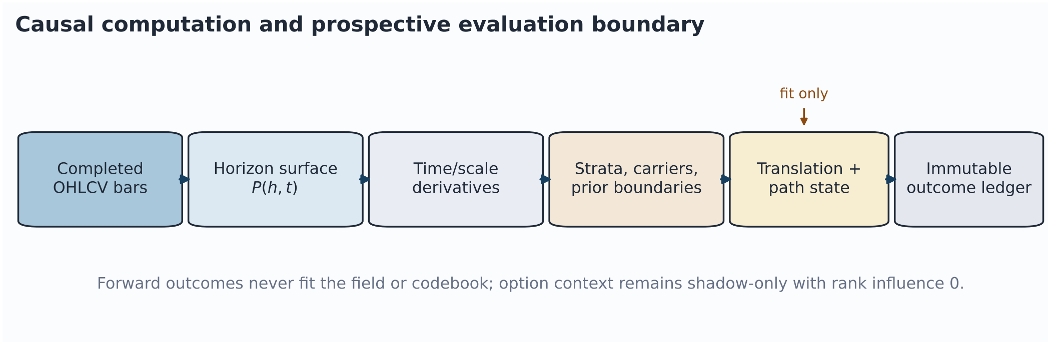

Figure 1: The auditable computation boundary. Audited experiments and option snapshots use completed bars; outcomes are recorded later and never refit the current field. The live research screen may include a provider's forming bar. A current field snapshot has zero direct rank and execution authority. Historical point-in-time field cohorts may enter a separately governed outcome-learning canary capped at 10%, and any display may influence a human.

3.2 Time/scale calculus and strata

For any series X, define a causal bounded difference

$$D_q(X)_t = \frac{r_t}{1+|r_t|}, \qquad r_t = \frac{\Delta X_t}{\mathrm{EWM}_q(|\Delta X|)_t}. \tag{3}$$

Successive applications form velocity $v = D_{13}(P)$, acceleration $a = D_{13}(v)$, jerk $j = D_{13}(a)$, and snap $s_4 = D_{13}(j)$. Numerical derivatives across $u_j = \log h_j$ yield scale gradient g, scale curvature c, and the bounded temporal change of scale gradient m. These are normalized finite differences; the kinematic vocabulary is mnemonic. Appendix A specifies the nonuniform-grid interior and edge stencils and every horizon reduction used to obtain the state vector.

The derivative surfaces are reduced into five defined strata. Structure combines signed-state strength and cross-horizon agreement. Kinematics weights $|v|, |a|, |j|, |s_4|$. Geometry combines $|g|$, $|c|$, $|m|$, and boundary energy. Information combines causal order-3 Bandt–Pompe entropy with an engineered disorder channel; its interpretation is explicitly limited to the fixed v1 window because sensitivity is material. Propagation summarizes co-occurring time and scale change, with signed cascade bias indicating orientation. Separate carriers report realized variation, volume participation, and an Amihud-like price-impact ratio (Amihud, 2002).

3.3 A calibration-relative language

The research interface aggregates five derivatives, seven strata, and three carriers into $\mathbf{x}_t \in \mathbb{R}^{15}$. Derivative, stratum, and carrier families each receive one third of total squared-distance weight. Robust center and scale are estimated only on the proper-fit segment. A deterministic farthest-point-initialized k-means procedure (MacQueen, 1967; Gonzalez, 1985) considers at most five Forms, requires each cluster to contain at least $\max(20, \lceil 0.05n \rceil)$ fit bars, and accepts a multi-Form codebook only when mean silhouette is at least 0.25 (Rousseeuw, 1987). Otherwise it emits one Form. The initialization is a deterministic variant, not MacQueen's original seeding rule; exact candidate selection and tie handling are given in Appendix C.

The later calibration segment provides state-conditional nearest-centroid distances. For evaluation distance d_t in state k,

$$\tau_t = \frac{1 + \#\{d_i^{\mathrm{cal},k} \geq d_t\}}{n_k + 1}. \tag{4}$$

The v1 payload's legacy key outside_learned_range is true when $\tau_t < 0.05$ and $n_k \geq 20$; throughout this paper we call that event the upper state-conditional calibration-distance tail. It does not imply coordinatewise range violation, convex-hull exclusion, density support, or a forecast. Equation 4 is a discrete empirical rank diagnostic: for $n_k = 20$ its minimum is $1/21$, so the strict 0.05 rule fires only beyond every same-state calibration distance. Form identifiers are request-local: F.001 for one symbol/window has no asserted identity elsewhere.

The visual translation layer shows the horizon cloud and three scopes: direction (P, v) colored by structure, structure–information colored by kinematics, and propagation–cascade bias colored by geometry. The second scope does not plot the legacy organization channel O. Display-only smoothing can soften the trace, but the raw path remains available. Loops are trajectories, not detected cycles.

4 Prospective Option Integration

The key design separates option mispricing from underlying path fit. Existing scanner eligibility and the champion opportunity score remain functions of cheapness, volatility edge, contract quality, execution quality, and a pre-existing recurrence term. A field snapshot is constructed only after an option already qualifies under the existing optionality logic. It cannot create eligibility and has zero direct weight in the champion score. A separate, versioned outcome-learning canary may blend historical closed-trade cohort evidence into the displayed score and order only after its evidence gates and a default-off operator-authorization gate pass; all families together are capped at 10%. The Market Field family can therefore have indirect historical-learning influence without granting the current field snapshot direct rank authority.

The original v1 contract interprets scanner candidates as long, directional, single-leg calls or puts: calls take side $r = +1$ and puts $r = -1$. Semantic 1.2 introduced, and 1.3 carries forward, the precedence signed net delta, explicit opening action plus option type, then the labeled legacy long-single-leg assumption; unsupported exposure abstains. Aggregate pressure and velocity are aligned as $p^\star = rp$ and $v^\star = rv$. The advisory state is mixed when $|p^\star| \leq 0.02$, contradictory when $p^\star < 0$, fading when $p^\star > 0$ and $v^\star < 0$, and supportive otherwise. Separately, the kinematic-exhaustion hypothesis uses pressure-aligned velocity $\widetilde{v} = \mathrm{sign}(p)v$. Four exploratory hypotheses—organized expansion, longward cascade, geometry/disorder shock, and kinematic exhaustion—use quantiles computed from prior rows only. The precise conditions appear in Appendix D.

New snapshots declare schema/model/revision, legacy shadow_only, zero direct rank influence, and no execution. Their authority receipt denies direct rank, veto, verdict, size, and execution power while declaring bounded downstream cohort use; retrospective Forms, outcomes, cross-market caches, and option inputs are excluded. Successful terminal-run finalization separately freezes ranks, scores, weight, versions, and a canonical hash; stale, pre-schema, and failed-finalization runs remain unsnapshotted. Authenticated ranking and candidate impressions begin at that independent schema boundary, so earlier exposure is unrecoverable. The canary remains an experiment, not evidence of option-selection value.

5 Diagnostic Evaluation

5.1 Questions, data, and protocol

We evaluate representation mechanics, not trading performance:

- RQ1 Is the serialized field prefix-invariant and the learned dictionary deterministic?
- RQ2 Do controlled paths produce the intended directional, organizational, and cross-horizon behavior?
- RQ3 Does the codebook decline weak separation, and how does its range diagnostic behave across assets?

RQ4 Does the multiscale vector show stronger unsupervised separation or window stability than a cheap single-horizon baseline?

RQ5 Does permutation-entropy window sensitivity propagate into SPY Form assignments and calibration-distance tails?

The primary audit uses adjusted daily Yahoo Finance OHLCV from 2018-01-01 through 2026-07-21 for SPY, QQQ, IWM, TLT, GLD, USO, VNQ, and BTC-USD. Each normalized local snapshot is hashed; the manifest records requested bounds, actual bounds, row count, adjustment mode, and SHA-256. The first-run manifest records cd67fb4 as its repository reference together with source-file hashes; it must not be read as the identity of later reruns. The current primary receipt records clean source HEAD 05e6433; the supplementary receipt records clean source HEAD 205fda0 and no evaluated source change during execution. Both execution heads contain implementation files byte-identical to the reviewed implementation commit f5d3884; the receipts bind those files and generated artifacts with SHA-256 hashes. The companion script can rerun exactly only when the local raw cache is retained, or retrieve a fresh provider snapshot whose hashes and results may differ. Provider terms prevent redistribution of the raw CSVs; the public package contains the manifest, hashes, derived results, and executed notebooks. Thus a hash identifies the original input but does not make that input recoverable from a fresh clone. A supplementary locally retained audit covers 15 datasets and all nine supported SPY timeframes, tests horizon-grid steps 1, 2, 4, and 8, and probes dictionary window stability (Appendix G).

For prefix invariance, we compare 11 channels across 19 horizons at four historical cut points per asset against the same timestamps computed with the complete suffix present. These are 32 prefix endpoints with 209 related, four-decimal serialized values at each endpoint, not 6,688 independent tests. Determinism reruns the full versioned lexicon and compares canonical JSON hashes. A deterministic 800-bar construction reverses log drift at bar 400; a second construction contrasts organized trend with alternating chop.

The matched baseline uses five causal 24-bar features: EMA12-minus-EMA24 normalized by ATR24, bounded EMA24 slope, signed path efficiency, realized-variation relative to causal EWM48, and mean volume relative to causal EWM48. It inherits each production run's requested warm-up (96 bars for the present grid), split indexes, proper-fit robust scaling, deterministic farthest-first k-means, support/silhouette gate, held-out state-conditional calibration distances, and evaluation chronology; its five coordinates receive equal weights. This controls the chronology and clustering procedure while changing the representation, but dimension and semantic content still differ. For entropy sensitivity, SPY is fully recomputed at trailing pattern-instance windows 8, 12, 24, 48, and 96; we report Information correlation, Form count, fit silhouette, label-invariant assignment agreement, transition rate, and calibration-tail rate. These are unsupervised diagnostics, not return labels.

5.2 Results

Across all 32 primary endpoints, the 6,688 serialized scalar comparisons are identical at 10^{-4} precision, with maximum observed serialized difference zero. The supplementary audit bypasses response rounding and passes 46/46 prefix checks spanning 24,472 full-precision values from live field matrices and aggregate derivatives, strata, carriers, and carrier ratios; maximum absolute error is zero at tolerance 10^{-12}. It does not yet exhaust dictionary assignments, minimum-input/initialization-coverage metadata, hypothesis flags, or every downstream option field. All eight tested primary lexicons reproduce byte-for-byte. On the synthetic reversal, threshold-crossing delays range from 10 to 28 bars over horizons 12 through 48, with Spearman $\rho = 1.000$. The organized segment has mean $|P| = 0.508$ and structure 0.859, versus 0.001 and 0.419 for alternating chop; permutation entropy increases from 0.385 to 0.568. The nonzero trend entropy reflects entropy of transformed pressure rows including startup and transition behavior, not entropy of a strictly monotone price sequence.

Under the chosen weighted Euclidean representation, k-means family, minimum-support rule, and silhouette threshold, multi-Form codebooks are accepted only for SPY (two) and USO (three). This may reflect weak separation, nonspherical structure, temporal drift,

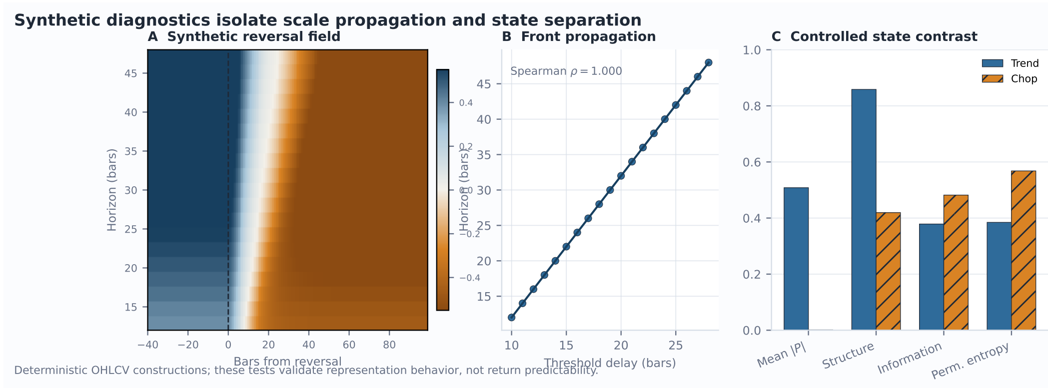

Figure 2: Controlled diagnostics. (A) A deterministic reversal moves through the horizon surface. (B) the threshold-crossing delay increases monotonically with horizon. (C) the intended trend/chop contrast appears in pressure magnitude, structure, information, and permutation entropy. These are construction tests, not return forecasts.

Diagnostic	Result	Interpretation
Prefix invariance	6,688 comparisons, 0 mismatches	No future-bar dependence detected at 10^{-4} API precision.
Deterministic dictionary	8/8 exact reruns	Fixed inputs yield byte-identical versioned lexicons.
Synthetic propagation	$\rho = 1.000$	Longer horizons cross the reversal threshold later by construction.
Distance-tail diagnostic	1.5–10.6% by asset	State-conditional empirical ranks are heterogeneous and non-exchangeable.
Codebook support gate	2/8 multi-Form	Multi-Form solutions appear only when the declared metric, support, and silhouette gates pass.
Compact snapshot	108.3 ms median; 4,395 bytes	Local compute-only benchmark; ranking and execution remain disabled.

Table 2: Representation and operational diagnostics. Exact-zero results apply at serialized precision; timing is a local compute-only benchmark.

limited support, or a conservative gate; it is not proof that other assets have one natural state. Among 7,237 evaluation bars with state-conditional support, 395 enter the upper 5% calibration-distance tail (5.46% pooled). Rates vary from 1.51% for VNQ to 10.58% for SPY (Figure 4); 33 USO evaluation bars lack the required same-state support and are reported separately from the denominator. Serial dependence, distribution shift, discrete ranks, and state conditioning prevent a nominal coverage interpretation. The shorter-window rates and Form instability in Appendix G are more consequential than the pooled proximity to 5%.

The cheap baseline is more readily partitioned by this gate: it accepts multiple Forms for 8/8 assets and has median fit silhouette 0.448, whereas Market Field accepts 2/8 and has median 0.000 because six one-Form fallbacks report zero. SPY window stability is mixed rather than dominant: Market Field has ARI 0.842 versus baseline 0.803 at the 70% prefix, but 0.751 versus 0.925 at 85%. This negative comparison rules out the claim that the present multiscale vector discovers more separated or consistently more stable unsupervised states than a cheap alternative. It remains useful as an explicitly decomposable measurement and visualization system, which is a different claim.

Entropy instability propagates selectively. On the same long-history SPY sample, alternative windows keep two Forms and assignment ARI remains 0.878–0.908 versus window 24, even though the aggregate Information correlation falls to 0.492 at window 8 and 0.511 at window 96. The upper calibration-distance-tail rate is less stable, ranging from 10.6% to 16.7%, and transitions range from 2.56 to 3.26 per 100 evaluation bars. Thus the coarse

Variant	Multi-Form	Silhouette	ARI	Tail (%)
Panel A: eight-asset representation comparison				
Market Field v1	2/8	0.000	0.84/0.75	5.5
24-bar baseline	8/8	0.448	0.80/0.92	4.6
Panel B: SPY entropy-window ablation				
H window 8	2	0.327	0.903	14.1
H window 12	2	0.329	0.908	12.3
H window 24	2	0.320	1.000	10.6
H window 48	2	0.322	0.878	16.7
H window 96	2	0.328	0.883	12.1

Table 3: Matched unsupervised diagnostics. Panel A reports multi-Form assets, median proper-fit silhouette, SPY assignment ARI at 70%/85% prefixes, and median asset tail rate. Panel B recomputes SPY while changing only the entropy window; ARI is against window 24. Silhouette is reported as zero when the gate returns one Form.

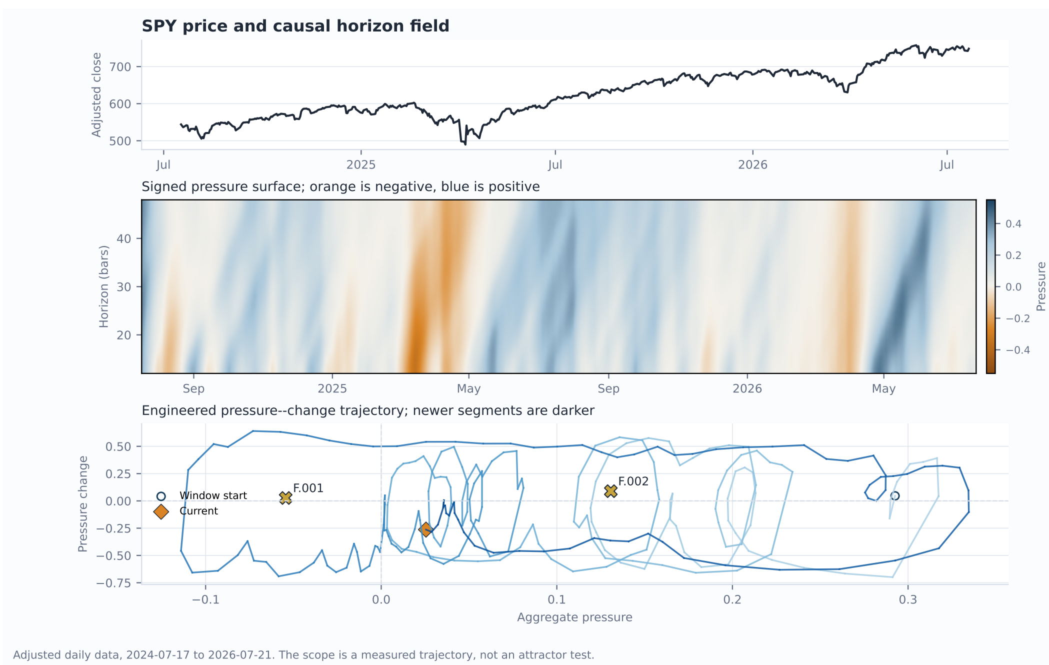

Figure 3: SPY example over the final 500 daily bars. Top: adjusted close. Middle: signed horizon field. Bottom: a pressure–change trajectory and request-local Form centroids. The phase portrait is an engineered projection, not an attractor test.

two-Form partition is comparatively robust here, but calibration-relative extremeness and transition frequency remain parameter-dependent. Figure 5 and row-level results are in the appendix and companion notebook.

Resolution sensitivity is material. Relative to a dense step-1 horizon grid over 8–64 bars, the supplementary step-4 audit has median correlation 0.940 but median IQR-normalized MAE 0.203 across 180 dataset–feature pairs; the 95th percentile error is 0.428. The reported total of 540 covers all three coarser grids (steps 2, 4, and 8), not step 4 alone. Pressure is comparatively stable, while geometric and propagation summaries move more. Because row-adjacent blending also changes with the grid, horizon density is a model parameter, not a rendering preference or pure numerical-resolution choice.

The run-specific primary compute and payload measurements appear in Table 2; the current supplementary distribution is preserved row-by-row in option_snapshot_latency.csv and linked to its source hashes by the run receipt. These volatile local measurements show feasibility for cache-only instrumentation, not a stable latency or payload contract. They exclude retrieval, persistence, fresh-process import, concurrent load, network and queueing, and a declared budget; live endpoint timing is a separate operational artifact. The emitted field payloads retain direct rank influence 0 and automatic execution false; the distinct outcome-learning receipt records any bounded indirect score effect.

6 Discussion and Limitations

The diagnostics support a systems claim: the live representation is non-anticipative on tested paths, deterministic for fixed inputs, responsive to controlled constructions, and instrumentable without direct field-snapshot or execution authority. The distinct outcome-learning canary is bounded and receipted but not validated as useful. The matched baseline is a deliberate negative result: Market Field does not show stronger unsupervised separation and is not established as a representation-learning advance. Its supported value is traceable multiscale measurement, visualization, and prospective experimental infrastructure.

Material limitations remain. Constants are hand designed, high-order differences amplify noise, and row smoothing changes with horizon density. Alternative entropy windows correlate only 0.342–0.630 in median with the 24-instance definition (minimum -0.014); SPY assignments remain coarse while tail and transition rates move. Ninety-six bars is availability, not convergence: median/p90 truncation error is 0.035/0.931 versus approximately 0/0.003 at 365. Semantic 1.3 separates finite startup values from full-dependency-support masks, but neither is convergence. Exchange-calendar completion and uniformly available volume remain absent.

The null state is nonintuitive: $Coh \in (7/27, 1]$, Structure is 0.42, and legacy O is 0.68 at a zero field; these encode agreement and low motion, not active organization. Forms cluster aggregate vectors rather than complete clouds and are window-dependent. Evaluation and outcome windows overlap; the option ledger omits full exposure, cost, exercise, capacity, and selection-bias controls. Cross-market vintages differ, association is not causation, and IV-minus-HV is neither a volatility surface nor arbitrage evidence. Pair v1 adds no efficacy evidence: its differences remain conditional on two request windows, shared support, provider aliases, currency and adjustment conventions, and compatible session labels. A field-coordinate gap is not relative performance or proof that one instrument leads another.

The development harness now executes purged rolling-origin comparisons (López de Prado, 2018), naive/EMA/technical, fixed two-window break, and fit-only two-state Gaussian-HMM comparators, raw-vector and family ablations, stationary-bootstrap intervals, and both false-discovery and family-wise corrections under a frozen machine-readable protocol. This closes an infrastructure gap, not the evidence gap: next work must freeze an untouched prospective holdout, add only preregistered comparators whose required inputs and conventions are defensible, and evaluate normalized option exposure with executable quotes, costs, and predeclared primary endpoints.

7 Conclusion

Market Field is an auditable measurement protocol, not a demonstrated representation-learning advance. Its cheap baseline separates more readily; prediction and the prospectively instrumented optionality hypothesis remain unproven.

Reproducibility Statement

Source, derived evidence, hashes, executed notebooks, environments, receipts, and the bibliography audit are included. Semantic 1.3 identifies the recipe, normalized input, and combined analysis but not provider truth. Restricted raw snapshots require an independently matching response; the README records commands and limits.

Ethics and Intended Use

This is exploratory research, not investment advice or autonomous trading. Displays and the bounded canary can influence order; promotion requires out-of-sample evidence and manual approval. Access control is not a research contribution.

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A Complete Field Calculus

A.1 Boundedness and prefix argument

For Equation 2, the triangle inequality bounds endpoint displacement by total path length. The map $x/(1+|x|)$ is bounded, prefix-initialized EWMs and the stated nonnegative row blends are convex, and past-only rolling windows and recursions preserve prefix measurability by induction in t. Clipping preserves the asserted unsigned bounds, while the declared zero-denominator cases close degenerate paths. This elementary argument establishes a property of the formulas, not causal inference; the endpoint audits test selected implementation paths.

A.2 Input contract and pressure construction

Experiments compute on retained histories before selecting the reported window. The interactive service may request additional past bars, but a provider can return fewer; response caps, exact prefetch rules, and transport behavior are operational contracts documented in the repository. Horizons are bar counts within one selected timeframe, and the system does not fuse timeframes into a single model.

Let $\epsilon = 10^{-9}$, let unqualified clip mean $\mathrm{clip}_{[0,1]}$, and define $\phi(x) = x/(1+|x|)$. The unsigned numerical clip first maps NaN to 0, $+\infty$ to 1, and $-\infty$ to 0. For consecutive finite observations, an EWM of span q follows

$$\mathrm{EWM}_q(x)_1 = x_1, \qquad \mathrm{EWM}_q(x)_t = (1-\alpha_q)\,\mathrm{EWM}_q(x)_{t-1} + \alpha_q x_t, \quad \alpha_q = \frac{2}{q+1}. \tag{5}$$

All EWMs use pandas adjust=False, ignore_na=False, and min_periods=0; the executed receipt pins the dependency version. Output is NaN until the first finite value, which initializes the EWM. A later missing value repeats the prior output but advances the absolute-position decay. If g missing positions follow an observed EWM y_s and x_{s+g+1} is finite, the next output is

$$y_{s+g+1} = \frac{(1-\alpha_q)^{g+1} y_s + \alpha_q x_{s+g+1}}{(1-\alpha_q)^{g+1} + \alpha_q}. \tag{6}$$

Thus Equation 5 is the $g = 0$ case; missing positions are not treated as zeros or simply removed from the time axis. The normalizer coerces timestamps and OHLC values, requires finite strictly positive OHLC with $H_t \geq \max(O_t, C_t, L_t)$ and $L_t \leq \min(O_t, C_t, H_t)$, drops invalid price rows, keeps the last duplicate timestamp, sorts rows, and reports every rejection category. Boundary comparisons accept only a tight isclose tolerance (relative and absolute tolerance 10^{-12}) to avoid rejecting floating-point boundary noise. Invalid or missing volume does not discard a valid price bar; it becomes unavailable for volume-dependent carriers and its coverage is reported. At least 60 valid price rows must remain for the public field. This visible input-quality contract is part of current semantic revision 1.3.

True range follows Wilder (J. Welles Wilder Jr., 1978):

$$TR_t = \begin{cases} H_t - L_t, & t = 1, \\ \max(H_t - L_t,\ |H_t - C_{t-1}|,\ |L_t - C_{t-1}|), & t > 1. \end{cases} \tag{7}$$

The true-range mean uses the available trailing $\min(h_j, t)$ observations. In Equation 1, $z_{j,t} = 0$ when that mean is zero. In Equation 2, $e_{j,t} = 0$ when the path sum is zero or M_{t-h_j} is unavailable. Otherwise the triangle inequality makes clipping mathematically redundant. Let $s_{j,t} = \mathrm{EWM}_5(d_{j,t}e_{j,t})$. The first adjacent-row blend is

$$q_{j,t} = 0.68 s_{j,t} + 0.32 N_j(s)_t, \tag{8}$$

where $N_1(s) = s_2$, $N_J(s) = s_{J-1}$, and $N_j(s) = (s_{j-1} + s_{j+1})/2$ in the interior (with $N_1(s) = s_1$ when $J = 1$). After $x_{j,t} = \mathrm{EWM}_3(q_j)_t$, the second blend is

$$P_{j,t} = 0.58 x_{j,t} + 0.42 \frac{x_{j-1,t} + 2x_{j,t} + x_{j+1,t}}{4}, \qquad 1 < j < J. \tag{9}$$

At an edge, the bracketed neighborhood becomes $(2x_{\text{edge}} + x_{\text{neighbor}})/3$; for $J = 1$, $P = x$. These convex operations preserve the pressure bound. Equal row adjacency, rather than distance in $\log h$, is the declared v1 smoothing model.

A.3 Base field channels

Write $u_j = \log h_j$ and let $\mathcal{G}_u$ denote the same nonuniform-grid derivative used by numpy.gradient(..., edge_order=1). If $J = 1$, the implementation sets both log-horizon derivative channels and the scaling exponent to zero. For $J \geq 2$, endpoints use one-sided slopes. For an interior row, with $a = u_j - u_{j-1}$ and $b = u_{j+1} - u_j$,

$$(\mathcal{G}_u X)_j = -\frac{bX_{j-1}}{a(a+b)} + \frac{(b-a)X_j}{ab} + \frac{aX_{j+1}}{b(a+b)}. \tag{10}$$

Let $R_1 = \mathcal{G}_u P$, $R_2 = \mathcal{G}_u R_1$, $G = 1.35|R_1|/(1 + 1.35|R_1|)$, $L = 1.35|R_2|/(1 + 1.35|R_2|)$, and $T = \mathrm{clip}(2.70|\Delta_t P|)$, where the first temporal difference is zero. Then

$$S = \mathrm{clip}(1.8|P|), \qquad Q = \mathrm{clip}(1.4|v|), \tag{11}$$

$$B = \mathrm{clip}(0.42G + 0.33T + 0.25L), \qquad Coh = \mathrm{clip}(1 - G/1.35), \tag{12}$$

$$E^{(0)} = \mathrm{clip}(0.60(1 - Coh) + 0.40Q), \qquad E = \mathrm{EWM}_4(E^{(0)}), \tag{13}$$

$$O = \mathrm{clip}(0.48Coh + 0.32S + 0.20(1 - E)). \tag{14}$$

Here S, Q, B, Coh, E, O are state strength, velocity strength, boundary energy, coherence analogue, disorder analogue, and the legacy display channel serialized as confidence. Because $G \in [0, 1)$, clipping is inactive in $Coh = 1 - G/1.35$ and $Coh \in (7/27, 1]$. Consequently the $0.42Coh$ term in Structure contributes strictly more than $49/450 \approx 0.109$; Structure cannot reach zero under v1. The state-vector stratum called Structure is different from O. Under the zero field, $(S, Q, G, L, T, E) = (0, 0, 0, 0, 0, 0)$, $Coh = 1$, and $O = 0.68$; this means agreement at zero and low disorder, not active organization. Further display channels are

$$\text{Expansion} = \mathrm{clip}(1.8\ \mathrm{sign}(P)v), \tag{15}$$

$$\text{Contraction} = \mathrm{clip}(-1.8\ \mathrm{sign}(P)v), \tag{16}$$

$$\text{Reflectivity} = \frac{\log(1 + 4\,\mathrm{clip}(0.50S + 0.22Q + 0.28B))}{\log 5}, \tag{17}$$

$$\text{Convection} = \mathrm{clip}\left(B(0.45 + 0.55Q)\right). \tag{18}$$

These names are operational. Coherence is not spectral coherence; disorder is not generally information entropy; reflectivity is not optical reflectivity; and convection does not assert fluid dynamics.

A.4 Research strata and carriers

For any row field Y, define signed normalization and causal difference operators

$$\mathcal{N}_q(Y)_{j,t} = \phi\left(\frac{Y_{j,t}}{\mathrm{EWM}_q(|Y_j|)_t}\right), \qquad \mathcal{D}_q(Y)_{j,t} = \mathcal{N}_q(Y_{j,t} - Y_{j,t-1}), \tag{19}$$

with first difference and result zero when the denominator is at most ϵ. Thus $v = \mathcal{D}_{13}(P)$, $a = \mathcal{D}_{13}(v)$, $j = \mathcal{D}_{13}(a)$, and $s_4 = \mathcal{D}_{13}(j)$. The research scale channels are

$$g = \mathcal{N}_{13}(\mathcal{G}_u P), \qquad c = \mathcal{N}_{13}(\mathcal{G}_u \mathcal{G}_u P), \qquad m = \mathcal{D}_{13}(g), \tag{20}$$

where every EWM normalizes along time, row by row. Thus m is a bounded temporal difference of the bounded scale gradient, not a raw mixed partial.

Order-3 permutation entropy is evaluated separately on each pressure row. At time t, a stable sort maps $(P_{j,t-2}, P_{j,t-1}, P_{j,t})$ to one of $3! = 6$ possible ordinal patterns, with earlier positions winning ties. The plug-in distribution uses the most recent at most 24 observed pattern instances:

$$H^{\mathrm{perm}}_{j,t} = -\frac{\sum_{\pi \in S_3} p_{\pi,j,t} \log p_{\pi,j,t}}{\log 6}, \tag{21}$$

with $0 \log 0 = 0$. Before two pattern instances exist, the value is zero. This short-window plug-in estimator is descriptive and can be downward biased; the material window sensitivity is reported in the main discussion. With bounded scale gradient g and velocity v,

$$c_{v,j,t} = \tanh\left(-\frac{v_{j,t} g_{j,t}}{g^2_{j,t} + 0.08}\right), \tag{22}$$

$$Pr_{j,t} = \mathrm{clip}\left(1.8\sqrt{|v_{j,t}||g_{j,t}|}(0.45 + 0.55 Coh_{j,t})\right). \tag{23}$$

The constant 0.08 is an engineered regularizer. For an idealized translating front $P(u,t) = F(t - \beta u)$, $P_t = F'$ and $P_u = -\beta F'$, so $-P_t P_u = \beta (F')^2$. This supports the sign convention—positive indicates motion toward larger horizons—but amplitude deformation can produce the same analogue.

The row-level strata are

$$\mathrm{Structure}_{j,t} = \mathrm{clip}(0.58 S_{j,t} + 0.42 Coh_{j,t}), \tag{24}$$
$$\mathrm{Kinematics}_{j,t} = \mathrm{clip}(0.18|v| + 0.24|a| + 0.27|j| + 0.31|s_4|)_{j,t}, \tag{25}$$
$$\mathrm{Geometry}_{j,t} = \mathrm{clip}(0.32|g| + 0.28|c| + 0.20|m| + 0.20B)_{j,t}, \tag{26}$$
$$\mathrm{Information}_{j,t} = \mathrm{clip}(0.72 H^{\mathrm{perm}}_{j,t} + 0.28 E_{j,t}). \tag{27}$$

Define horizon-weighted and arithmetic reductions

$$\mathcal{A}_h(Y)_t = \frac{\sum_{j=1}^J h_j Y_{j,t}}{\sum_{j=1}^J h_j}, \qquad \mathcal{A}_0(Y)_t = \frac{1}{J}\sum_{j=1}^J Y_{j,t}. \tag{28}$$

The five derivative coordinates use $\mathcal{A}_h$. Structure, Kinematics, Geometry, Information, and Propagation use $\mathcal{A}_0$, with

$$\mathrm{Propagation}_t = \mathcal{A}_0(Pr)_t, \qquad \mathrm{CascadeBias}_t = \begin{cases} \dfrac{\sum_j Pr_{j,t} c_{v,j,t}}{\sum_j Pr_{j,t}}, & \sum_j Pr_{j,t} > \epsilon, \\ 0, & \text{otherwise.} \end{cases} \tag{29}$$

Let $C_t^+ = \max(C_t, \epsilon)$ and $r_t = \Delta \log C_t^+$ with $r_1 = 0$. Realized variation is $RV_{j,t} = \sqrt{\sum_{i=t-h_j+1}^{t} r_i^2}$ over available observations, without annualization. The local scaling exponent is

$$\gamma_{j,t} = \mathrm{clip}_{[-2,2]}\left[\mathcal{G}_u \log(RV + \epsilon)\right]_{j,t}, \qquad \mathrm{ScalingExponent}_t = \mathrm{clip}_{[-2,2]}\, \mathcal{A}_0(\gamma)_t. \tag{30}$$

It is not a Hurst exponent or multifractal spectrum. For equal per-bar squared variation, $RV_h \propto h^{1/2}$ and $\gamma \approx 0.5$ is the natural reference. A value near 0 means accumulated variation changes little across the local scale neighborhood (for example, one common shock dominates), while 1 means approximately linear growth in RV_h. Because nested-window $RV_{j,t}$ is nondecreasing in h_j, $f_j = \log(RV_{j,t} + \epsilon)$ is nondecreasing. Endpoint one-sided secants are therefore nonnegative, and the interior stencil in Equation 10 equals

$$\frac{b}{a+b}\frac{f_j - f_{j-1}}{a} + \frac{a}{a+b}\frac{f_{j+1} - f_j}{b} \geq 0. \tag{31}$$

Thus $\gamma \geq 0$ in exact arithmetic. The symmetric lower clip is only a defensive implementation guard: a materially negative preclip or emitted value is a numerical, unsorted-grid, inconsistent-window, or implementation-quality failure to investigate, not an interpretable state or evidence of anti-persistence. The service retains such a value only in its invalid diagnostic channel, withholds it from option translation, and suppresses the request's Form dictionary and retrospective relationship atlas so the impossible coordinate cannot influence a learned output.

Let $\widetilde{V}_t = V_t$ when volume is finite and nonnegative, and let it be unavailable otherwise. Define the pointwise impact observation

$$Z_t = \begin{cases} |r_t|/(C_t^+ \widetilde{V}_t), & \widetilde{V}_t > 0, \\ \text{NaN}, & \text{otherwise}, \end{cases} \tag{32}$$

so missing, invalid, and zero volume make that impact observation unavailable rather than zero. The row inputs

$$U_{j,t} = \text{mean}_{i=t-h_j+1}^{t,\, \widetilde{V}_i \text{ available}} \widetilde{V}_i, \qquad I_{j,t} = \text{mean}_{i=t-h_j+1}^{t,\, Z_i \text{ available}} Z_i \tag{33}$$

use available observations only (pandas rolling means with min_periods=1); either mean is unavailable when its window has no admissible observation. Here I is Amihud-like (Amihud, 2002). For $X \in \{RV, U, I\}$ and $b = \max(34, 2h_J)$, the bounded carrier field and separately displayed direct ratio are

$$L_b(X)_{j,t} = \text{clip}\left[0.5 + 0.5\tanh\left(2\log\frac{X_{j,t} + \epsilon}{\text{EWM}_b(X_j)_t + \epsilon}\right)\right], \tag{34}$$

$$R_b(X)_{j,t} = \text{clip}_{[0,10]}\,\frac{X_{j,t} + \epsilon}{\text{EWM}_b(X_j)_t + \epsilon}. \tag{35}$$

The three state-vector carriers are $\mathcal{A}_0(L_b(RV))$, $\mathcal{A}_0(L_b(U))$, and $\mathcal{A}_0(L_b(I))$. The pandas EWM baseline is evaluated on the masked series and includes the current available observation. A direct ratio is unavailable wherever its current rolling input is unavailable; participation and impact ratios also require at least one positive-volume observation in the input. Missing internal carrier cells are filled with 0.5 only after this calculation to keep the optional retrospective clustering path finite: 0.5 is a neutral computational placeholder, not an observation, and must not be reported as a measured 1.0× ratio. Impact cannot be observed from a zero-volume bar.

B Fixed Constants and Rationale

Table 4 inventories the principal hand-set constants. They were inherited from the deployed v1 design and fixed before the baseline and entropy analyses in this revision; that chronology prevents tuning to the new results but is not preregistration or evidence of optimality.

C Dictionary Chronology and Semantics

Using Equation 28, the exact state vector is

$$\begin{aligned}\mathbf{x}_t = (&\mathcal{A}_h(P), \mathcal{A}_h(v), \mathcal{A}_h(a), \\ &\mathcal{A}_h(j), \mathcal{A}_h(s_4), \mathcal{A}_0(\text{Structure}), \mathcal{A}_0(\text{Kinematics}), \\ &\mathcal{A}_0(\text{Geometry}), \mathcal{A}_0(\text{Information}), \text{Propagation}, \text{CascadeBias}, \\ &\text{ScalingExponent}, \mathcal{A}_0(L_b(RV)), \mathcal{A}_0(L_b(U)), \mathcal{A}_0(L_b(I))) \in \mathbb{R}^{15}.\end{aligned} \tag{36}$$

Component	Constants	Design rationale / status
Pressure time smoothing	EWM spans 5, 3	Short causal denoising stages inherited from v1.
Pressure row blends	0.68/0.32; 0.58/0.42	Preserve own-row dominance while sharing adjacent scale context.
Base channel gains	1.35, 1.4, 1.8, 2.70	Map bounded visual analogues into useful display ranges; uncalibrated.
Motion normalization	EWM span 13	Common causal scale for successive bounded differences.
Structure	0.58 activity, 0.42 agreement	Hand-set composite; creates the disclosed nonzero floor.
Kinematics	0.18/0.24/0.27/0.31	Increasing weight on higher-order motion; not estimated.
Geometry	0.32/0.28/0.20/0.20	Balances gradient, curvature, mixed change, and boundary energy.
Information	0.72 entropy, 0.28 disorder	Makes ordinal entropy load-bearing; sensitivity is explicitly tested.
Propagation	0.08 regularizer; 1.8 gain; 0.45/0.55 agreement blend	Stabilizes the level-set analogue and emphasizes agreement.
Carriers	$b = \max(34, 2h_J)$; ratio cap 10	Longer causal reference than the largest configured horizon.
Metric families	1/3 each	Equal total weight for derivatives, transforms, and carriers.
Dictionary gate	$k \leq 5$; max(20, 5%) support; silhouette 0.25	Conservative deterministic acceptance rule.
Signature	0.35 bins	Coarse request-local identity; no semantic meaning.
Option hypotheses	20-bar minimum; quantiles 35–75%	Exploratory prior-row thresholds, not fitted classifiers.
Scanner base	0.30/0.28/0.27/0.15; recurrence 0.12	Pre-existing option heuristic; the current Market Field snapshot's direct base-score weight remains zero.

Table 4: Principal fixed constants. No listed value is inferred from the baseline or entropy-sensitivity outcomes.

The first five, next seven, and final three coordinates form derivative, field-transform, and OHLCV-carrier families.

For N available rows, v1 uses zero-based split indexes

$$t_E = \max(5, \min(\lfloor 0.60N \rfloor, N-6)), \tag{37}$$

$$t_F = \min(w, \max(0, t_E - 40)), \qquad w = \max(34, 2h_J), \tag{38}$$

$$n_{cal} = \min\{\max(20, \lfloor (t_E - t_F)/3 \rfloor), \max(0, t_E - t_F - 20)\}, \tag{39}$$

$$t_C = t_E - n_{cal}. \tag{40}$$

Proper fit is $[t_F, t_C)$, held-out distance calibration is $[t_C, t_E)$, and evaluation is $[t_E, N)$. Thus the fit and calibration segments retain at least 20 rows when the input guards permit. Scaling, codebook selection, centroids, and transition grammar use proper fit only; state-conditional distance references use calibration; evaluation supplies displayed sequences and descriptive forward annotations.

For feature ℓ, proper-fit center is the median and initial scale is the interquartile range computed by NumPy quantiles with method=linear. If scale is at most ϵ, fallback order is 1.4826 times median absolute deviation, population standard deviation, and finally 1, each selected only when greater than ϵ. Let $z_{t\ell} = (x_{t\ell} - m_\ell)/s_\ell$. Squared metric distance is

$$d(\mathbf{x}_t, \boldsymbol{\mu})^2 = \frac{1}{15}\sum_{\ell \in \mathcal{D}}(z_{t\ell} - \mu_\ell)^2 + \frac{1}{21}\sum_{\ell \in \mathcal{S}}(z_{t\ell} - \mu_\ell)^2 + \frac{1}{9}\sum_{\ell \in \mathcal{C}}(z_{t\ell} - \mu_\ell)^2, \tag{41}$$

where family sizes are 5, 7, and 3. Equivalently, coordinates are multiplied by the square root of their displayed weight before Euclidean k-means and silhouette computation.

For each candidate $k = 2, \dots, \min(5, \lfloor n_{fit}/n_{min} \rfloor)$, where $n_{min} = \max(20, \lceil 0.05 n_{fit} \rceil)$, the first seed is the fit point closest to the metric-space origin. Each later seed is the point farthest from its nearest existing seed; first row wins an exact tie. Lloyd updates run for at most 40 iterations, nearest-centroid ties choose the lowest index, and an empty cluster retains its previous centroid. Convergence requires unchanged assignments and centroids within absolute tolerance 10^{-12} and zero relative tolerance. The signature coordinate is $q_\ell = \text{np.rint}(\mu_\ell/0.35)$ cast to little-endian 32-bit integer; rint rounds to nearest integer with exact half ties to even. A candidate is discarded if any cluster lacks n_{min} rows, any centroid separation is at most 10^{-6}, two centroids share the same signature vector, or mean metric-space silhouette is below 0.25. The accepted candidate maximizes silhouette, with smaller k winning a tie; if none passes, $k = 1$. Final centroids are ordered lexicographically after rounding metric coordinates to 10 decimals. These choices are deterministic engineering rules, not evidence that the resulting Forms are natural latent states.

The reported legacy match transform uses the median of pooled calibration nearest-centroid distances (falling back to fit distances only if calibration is empty):

$$\text{match}_t = \exp\left(-d_t / \max(\text{median}(d^{\text{cal}}), \epsilon)\right). \tag{42}$$

It is a monotone resonance transform, not a probability; the state-conditional empirical tail rank in Equation 4 is the more transparent distance context. Evaluation outcomes use overlapping windows and are not independent. Form names summarize calibration-relative measurement profiles; they are not universal economic regimes or concepts learned independently of the feature design.

Asset	Bars	Forms	Silhouette	Tail n	Outside (%)
SPY	2,148	2	0.320	860	10.6
QQQ	2,148	1	0.000	860	4.4
IWM	2,148	1	0.000	860	6.6
TLT	2,148	1	0.000	860	3.3
GLD	2,148	1	0.000	860	6.6
USO	2,148	3	0.315	827	6.8
VNQ	2,148	1	0.000	860	1.5
BTC-USD	3,124	1	0.000	1,250	4.4
Pooled	18,160	–	–	7,237	5.5

Table 5: Daily-series dictionary diagnostics. Silhouette is zero when the support gate returns one Form. Tail n counts evaluation bars with at least 20 state-conditional calibration references. The table's "Outside" column preserves the v1 legacy label and means upper 5% calibration-distance tail.

D Completed-Bar Option Contract

The option snapshot retains at most 365 daily bars, requires at least 96, and treats a session as complete after 16:15 America/New_York. It uses horizons $\{12, 14, \dots, 48\}$ and excludes retrospective dictionary sections. The 96-bar threshold is the disclosed $2h_{\max}$ initialization target adopted after the truncation audit; snapshots with 60–95 bars are unavailable and report how many bars are still needed. Available snapshots also record alignment basis, input quality, authority, and a 15-row coordinate-coverage block. Each row separates a finite computed value from the per-time full-dependency-support mask and reports required inputs, the declared support lower bound, retained-prefix depth, current availability, and whether a finite internal neutral placeholder stands in for an unavailable carrier. Unavailable option responses carry canonical bars-needed metadata but no fabricated 15-row analysis block. maturity remains only a legacy serialized alias. These masks do not make the infinite-memory EWM identical to a full-history calculation. Support and resistance are strictly shifted:

$$\text{Support}_t = \min_{i=t-20}^{t-1} L_i, \qquad \text{Resistance}_t = \max_{i=t-20}^{t-1} H_i. \tag{43}$$

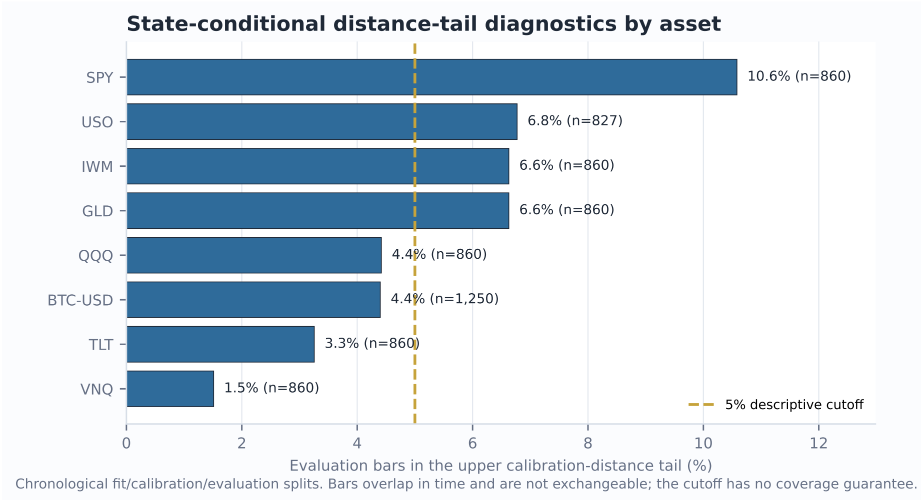

Figure 4: Asset heterogeneity in the empirical distance-tail diagnostic. The dashed line is a descriptive cutoff, not a promised coverage level.

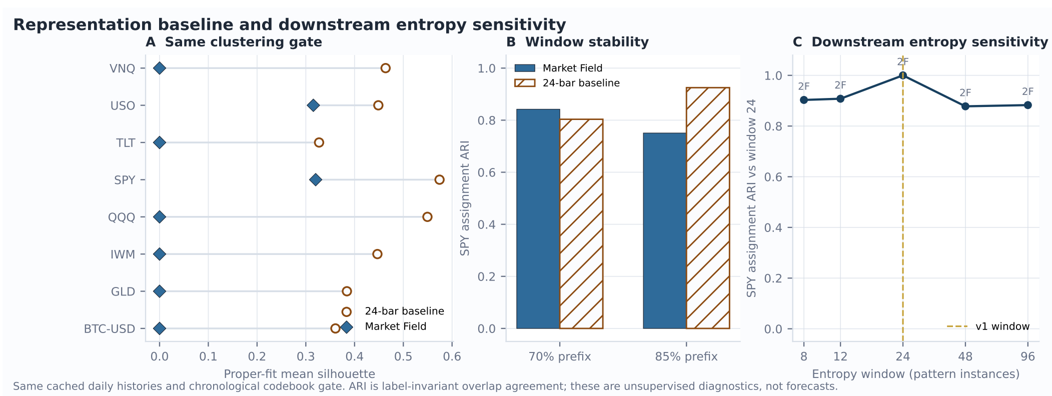

Figure 5: Matched representation and entropy diagnostics. (A) Proper-fit silhouette under the same codebook gate. (B) SPY partition agreement between prefix-fitted and full-window assignments. (C) SPY assignment agreement after recomputing the complete dictionary with alternative entropy windows; annotations show Form count. These panels assess unsupervised construction behavior, not prediction.

ATR is a trailing 14-bar simple mean of true range. Range position, close-to-SMA20, five-bar return, and ATR distance to the boundaries provide price-action context.

All hypothesis thresholds are computed from rows strictly before the current row and require at least 20 finite observations. Quantiles use NumPy's default linear interpolation:

- Organized expansion: structure $\geq q_{70}$, propagation $\geq q_{60}$, information $\leq q_{50}$, and $|p| \geq q_{60}$.
- Longward cascade: propagation $\geq q_{72}$, cascade bias $\geq \max(0.08, q_{62})$, and $|p| \geq q_{55}$.
- Geometry/disorder shock: geometry $\geq q_{75}$ and information $\geq q_{60}$.
- Kinematic exhaustion: kinematics $\geq q_{75}$, $|p| \geq q_{65}$, and pressure-aligned velocity $\widetilde{v} = \mathrm{sign}(p)v \leq q_{35}$.

These are deterministic exploratory definitions, not trained classifiers. The legacy fallback assumes a long directional single-leg interpretation; semantic v1.3 instead prefers signed delta or opening action plus option type, and abstains for unsupported exposure rather than applying that fallback to a known short or multi-leg structure.

Each scanner component is an existing heuristic on $[0, 100]$. Its base score remains

$$B = 0.30\,\text{cheapness} + 0.28\,\text{volEdge} + 0.27\,\text{contractQuality} + 0.15\,\text{executionQuality}. \tag{44}$$

For nonnegative integer recent-symbol, total-symbol, and recent-group counts (s_r, s_T, g_r), the pre-existing recurrence component is

$$\begin{aligned} R = \operatorname{clip}_{[0,100]}(&18 \min(s_r, 4) + 4 \min(\max(s_T - 1, 0), 5) \\ &+ 3 \min(\max(g_r - s_r, 0), 6)), \qquad \text{rank} = \operatorname{clip}_{[0,100]}(B + 0.12R). \end{aligned} \tag{45}$$

Market Field has zero direct weight in both equations. Separately, let $\ell_f \in [0, 100]$ be a reliability-weighted actual-close cohort score for available learning family f, including recurrence, replacement, Market Field, contract direction, and duration. A family is available only when the candidate cohort and at least one comparison cohort each contain at least eight closes. With normalized reliability weights α_f and mean reliability $\bar{\rho}$, the learning score and maximum proposed blend are

$$L = \sum_f \alpha_f \ell_f, \qquad w = 0.10 \min(1, N/40) \min(1, q/0.10)\bar{\rho}, \tag{46}$$

where N is the count labeled as classified actual-close cycles and q is their non-weak-process share. The applied score is $A = (1 - w)\,\text{rank} + wL$ only when $N \geq 40$, $q \geq 0.10$, at least one family is comparable, and the bounded canary is explicitly enabled by operator configuration; otherwise $A = \text{rank}$ while the counterfactual remains observable. Thus 10% is a hard total cap, not the Market Field family's weight. The policy cap cannot change automatically, although the event-level applied weight scales deterministically with the frozen evidence. The receipt decomposes each family's score contribution and leave-one-component-out rank effect.

In the manager, current field evidence may lower displayed advisory confidence or increase displayed urgency; because the decision window is rebuilt after that advisory change, urgency may also recompute next_review_date. It cannot change hard vetoes, verdicts, targets, sizing, or execution. The challenger remains unpromoted until at least 100 cycles, 25 additional new cycles, incremental out-of-sample value, and manual approval. Separate metamorphic tests compare supportive and contradictory contexts at the scanner/selection boundary and across stored-position, manager, and manual lifecycle-recording boundaries; within those tested boundaries, eligibility, selected contract, hard vetoes, target size, quantities, prices, P&L, and execution authority remain invariant. This is not one continuous scanner-to-broker proof: the lifecycle test seeds the persisted event and contract, and the application does not own an automated broker execution path. Successfully finalized rank snapshots and authenticated browser impressions are append-only and idempotent from the independent rank/exposure-schema deployment boundary.

E Pairwise Relative Field Receipt

Relative Field Pair v1 compares a target A and benchmark B only after each has been built independently under the same formula revision, timeframe, horizon grid, and settings. Let $X^A_{j,t}$ and $X^B_{j,t}$ be coordinate j of the two 15-dimensional vectors and let $M^A_{j,t}, M^B_{j,t}$ denote their source-observed, full-dependency-support masks. On an admissible aligned timestamp,

$$\Delta^{\text{native}}_{j,t} = X^A_{j,t} - X^B_{j,t}, \qquad M^A_{j,t} = M^B_{j,t} = 1. \tag{47}$$

Missing, provisional, invalid, or neutral-placeholder evidence is not coerced to zero. The sign means target minus benchmark in that coordinate's implemented 15-dimensional scale, not raw market units; it is not a utility ordering. In particular, the three carrier coordinates are bounded causal-baseline relative levels rather than raw realized variation, volume, or impact. Higher disorder, propagation, realized-variation state, participation, or liquidity stress is not inherently better.

Each component dictionary retains its own proper-fit location m_j^q and robust scale s_j^q, $q \in \{A, B\}$. Pair context is

$$\Delta_{j,t}^{\text{context}} = \frac{X_{j,t}^A - m_j^A}{s_j^A} - \frac{X_{j,t}^B - m_j^B}{s_j^B}. \tag{48}$$

It is emitted only on the intersection of the two evaluation segments, after both fixed proper-fit and held-out calibration partitions. Fit or calibration bars are not retrospectively displayed with those later reference statistics, and an unavailable scale remains unavailable. Thus the displayed context trace uses no future observation relative to its evaluation bar, but it remains request-window-relative: changing either retained history can change the two references. Request-local Form IDs and centroids are never matched across symbols.

Price progress is separate from field geometry. It uses the full-precision normalized close path carried beside each component analysis rather than the four-decimal display serialization. If p_0^A, p_0^B are the first common aligned closes, the displayed relative index is

$$RI_t = 100\frac{p_t^A/p_0^A}{p_t^B/p_0^B}. \tag{49}$$

Active progress is $RI_t - 100$. For the separate beta-adjusted path, let $r_t^q = \log(p_t^q/p_{t-1}^q)$. Once at least 20 prior aligned return pairs exist, $\widehat{\beta}_{t-1}$ is the centered covariance/variance slope—equivalent to the slope from an intercept-inclusive fit—over at most the 60 pairs strictly before t. The displayed increment is $r_t^A - \widehat{\beta}_{t-1} r_t^B$; it does not subtract the fitted intercept and is therefore not an OLS abnormal-return residual or alpha. The estimate is unavailable if the benchmark population return standard deviation is below 10^{-7}, if the result is nonfinite, or if $|\widehat{\beta}_{t-1}| > 25$. An invalid estimate is rejected rather than clipped or carried forward: the row's beta-adjusted return is unavailable and the cumulative chain resets, with any later valid estimate starting a new chain. The current beta/return summary uses the latest aligned row only and is unavailable whenever its beta is unavailable. The displayed cumulative beta-adjusted return is $100\{\exp[\sum_u (r_u^A - \widehat{\beta}_{u-1} r_u^B)] - 1\}$ within each contiguous available chain. It is not used to transform field coordinates. Neither RI_t nor a beta-adjusted return changes a Form or supplies a field "winner."

The alignment receipt declares its rule, per-leg dropped-observation counts at or after the displayed window start, and separately identified unmatched-tail counts. Those tails are subsets of the dropped counts; pre-window truncation is not counted as pair dropout. Daily and weekly rows align by serialized market-session date; the service does not independently certify exchange calendars or timezones. When intraday source timestamps retain timezone information, both sides normalize to UTC and require exact equality. Timezone-naive provider/cache rows instead require exact equality of their serialized naive timestamps and are not relabeled as UTC. Neither branch uses a nearest-neighbor join or forward fill, and matching timestamps do not certify common sessions: nonidentity pairs retain unknown session compatibility. Nonidentity DXY comparisons at 1h, 2h, and 4h are explicitly unavailable under the current provider anchors; DXY/DXY remains an algebraic identity control. The live endpoint uses provider/cache rows as returned and does not independently certify its latest bar as exchange-complete. The canonical DXY selector resolves to Yahoo's DX-Y.NYB index identifier and retains that alias in provenance; UUP is not substituted. Provider, adjustment, currency, timezone, and session differences remain comparison limitations.

The ordered comparison hash binds the target analysis hash, benchmark analysis hash, alignment contract, and normalization contract. Swapping A and B changes the receipt and reverses signed differences. The compact fit-relative stretch summary takes the same supported-coordinate intersection at t and $t-5$, averages absolute context gaps within each of the three coordinate families, and then averages the family means. It is unavailable when any family lacks common support; a change no larger than $\max(0.05, 0.05S_{t-5})$ is labeled mixed. Each scope selector shows target, benchmark, or difference on one axis domain shared across all three subjects; differential coordinates are centered at zero. Their paths are visual projections, not detected cycles, attractors, lead–lag estimates, or connectedness.

Pair v1 has zero scanner, canary, veto, verdict, sizing, or execution authority. Basket fields, cross-sectional peer ranks, persistent cross-symbol Forms, cross-timeframe fusion, predictive value, and economic value remain future work. The present paper adds this auditable implementation boundary but no Pair-v1 efficacy result. A standalone reconstructibility supplement, relative-field-addendum.tex, specifies the exact alignment, precision, beta-adjustment, stretch, receipt, and failure contracts.

F Cross-Market Shadow Atlas

The research page can read cached energy, real estate, agriculture, metals, sector-divergence, and crypto measures. For each source, it screens Spearman association (Spearman, 1904) between source pressure change and daily target log return at lags 0, 1, 5, and 20. The first 70% selects the largest absolute association. Holdout inference uses 199 fixed-block permutations of ranked source values with block length five and Benjamini–Hochberg correction (Benjamini & Hochberg, 1995). This controls false discovery rate under its assumptions, not family-wise error rate. A relationship is called persistent only when calibration and holdout signs agree, holdout $|\rho| \geq 0.15$, and $q \leq 0.10$.

This atlas is explicitly shadow_only with no field influence. It is association screening, not evidence of a spillover mechanism. Sources have heterogeneous cache vintages and are daily even when the selected field is intraday. The option snapshot intentionally excludes them until stronger alignment, provenance, and prospective validation exist.

G Multi-Timeframe and Resolution Sensitivity

The supplementary locally retained snapshot contains 9,235 completed bars across 15 datasets: SPY at 1m, 5m, 15m, 30m, 1h, 2h, 4h, 1D, and 1W, plus six additional daily markets. It excludes 15 latest rows under a conservative completion policy and finds zero duplicate timestamps, missing OHLCV cells, or invalid OHLC rows. All 46 prefix checks, including a direct mutation of the unseen SPY suffix, have zero error across 24,472 full-precision numeric values at tolerance 10^{-12}; all 16 repeated canonical-hash checks match. The audit covers live matrices and aggregate derivatives, strata, carriers, and ratios while bypassing response rounding. Every evaluated channel is finite across all nine timeframes. These are availability and implementation-invariance results, not evidence that one parameterization transfers economically across sampling frequencies.

For grid sensitivity, step 1 over horizons 8–64 is the reference and steps 2, 4, and 8 are recomputed from the same completed prefixes. The audit discards $W = \min(128, \max(60, \lfloor N/3 \rfloor))$ startup rows. For reference series y and candidate $\widehat{y}$, it reports

$$\mathrm{NMAE}_{IQR} = \frac{\mathrm{mean}_t\, |y_t - \widehat{y}_t|}{Q_{0.75}(y) - Q_{0.25}(y)}, \tag{50}$$

where $Q_{0.25}$ and $Q_{0.75}$ use NumPy method=linear on pairwise-finite reference values; the statistic is unavailable when their difference is at most 10^{-9}. Grid and entropy-window correlations are Pearson product-moment coefficients from numpy.corrcoef after pairwise deletion of nonfinite values. A correlation is unavailable with fewer than three pairs or when either population standard deviation is at most 10^{-12}; aggregate medians and percentiles omit unavailable values and use linear interpolation. Across 15 datasets and 12 aggregate features, step 4 contains 180 pairs and produces median correlation 0.940 and median NMAE_{IQR} 0.203; its 95th-percentile error is 0.428. All three coarser grids together contain 540 pairs. The worst step-4 pair is the SPY 15m scaling exponent (NMAE_{IQR} 0.730, correlation 0.911). This does not define an optimal grid. It demonstrates that convergence must be measured per channel and that denser horizon resolution cannot be described as a purely visual enhancement.

Dictionary stability depends on sample length. On the 749-bar supplementary SPY daily window, the 70% prefix selects one Form while the full window selects two, producing adjusted Rand index (Hubert & Arabie, 1985) 0.000 on their overlap; the 85% prefix and full window both select two and achieve 0.873. On the longer 2,150-bar primary SPY history, both prefixes retain two Forms but ARI is 0.842 at 70% and 0.751 at 85%. The matched single-horizon baseline yields 0.803 and 0.925. Shorter-window SPY and QQQ upper calibration-distance-tail rates are 36.7% and 42.7%, versus 10.6% and 4.4% in the longer primary sample. These differences are direct evidence that Form identity and empirical distance ranks are window- and distribution-dependent. Only two next-state entries meet the current reliability support across seven supplementary daily dictionaries.

A live operational probe artifact records HTTP 200 for all nine timeframe selectors and its run-specific endpoint timings, but conservatively flagged all latest provider bars as possibly forming. Consequently, paper results use locally retained completed prefixes. Live endpoint availability and timing are reported only as an operational check, not a stable performance distribution.

H Evaluation Details and Threats to Validity

Synthetic construction. The deterministic reversal series contains 800 bars, a positive log drift before bar 400 and a negative drift afterward, with fixed OHLC ranges and volume. Threshold delay is the first post-reversal bar where $P_{h,t} < -0.05$. The trend/chop contrast holds bar ranges and deterministic generation fixed while changing directional path organization. These tests are useful precisely because the expected behavior follows from construction; they do not estimate generalization.

Prefix audit. For each of eight daily series and four cut points, the primary script rebuilds both the prefix and full history. It compares 11 four-decimal serialized channels over 19 horizon rows at the matching final prefix timestamp. The supplementary full-precision audit expands live numeric coverage as described above. Zero difference means the tested paths did not use the later suffix; it does not prove causal inference, provider timestamp correctness, or absence of bugs outside those paths. Remaining regression scope includes minimum-input/initialization-coverage metadata, hypothesis flags, lexicon fields under a fixed split, and the complete final option snapshot.

Distance tails. The pooled rate in Table 2 is descriptive. Bars are serially dependent; calibration sizes vary by state; the same hyperparameters were developed while observing related markets; and distributions drift. Asset-level dispersion in Figure 4 is more informative than the pooled proximity to 5%.

Operational benchmark. Timing uses one local Windows process with imports already loaded and cached in-memory histories. The supplementary distribution contains 63 warm runs across seven symbols; it excludes retrieval, persistence, fresh-process import time, concurrent load, network, and production queueing. It cannot be compared directly to production request latency or a service-level objective.

Data quality. Yahoo Finance adjusted OHLCV is convenient and reproducible only up to provider revisions. It is not the live IBKR path used by every application request. Corporate-action adjustments alter historical levels. The package therefore records canonical row hashes and never presents the sample as immutable exchange truth. Raw snapshots are retained locally but not redistributed; a fresh provider retrieval is a new sample even when its requested ticker and dates match.

I Research Progression and Design Rationale

1. Horizon strips. The starting point applied a common indicator across lookbacks, preserving SwamiCharts' visual scan across scale.
2. Continuous pressure. Indicator votes were replaced with a signed, efficiency-weighted displacement so direction and organization remained continuous.
3. Causal derivatives. Pressure change and successive bounded differences exposed strengthening, fading, and high-order instability without future bars.

4. Scale geometry and information. Derivatives over log horizon, boundary energy, and ordinal entropy separated across-scale fragmentation from within-scale motion.
5. Field cloud and scopes. A narrow horizon cloud became the overview; phase projections moved detailed relationships out of linear metric cards.
6. Translation layer. A deterministic request-local codebook attached compact names to measured profiles while exposing support and range evidence.
7. Context atlas. Prior-bar boundaries, option snapshots, and cached cross-market associations were added as visibly separate evidence layers.
8. Relative Field Pair. Two independently computed same-recipe fields can be compared through shared-support coordinate differences, relative price context, and common-axis scopes without matching request-local Forms or granting the pair decision authority.
9. Prospective options instrumentation. Immutable snapshots entered scanner and manager workflows with zero direct field weight, bounded indirect learning, and no execution authority, creating a path to future evaluation without silently changing eligibility.

This progression is a design synthesis of established ideas, not a claim of new market physics. Its most important property is traceability: a phrase in the interface can be mapped back to a versioned vector, a completed data prefix, and explicit transformations.

J Economic-Evaluation Protocol and Remaining Evidence

The repository now includes a machine-readable development protocol and deterministic prequential runner. It freezes chronological origins, fit/calibration/evaluation windows, purge/embargo, endpoints, methods, and a trial registry; reports unsupported rows separately; compares naive, EMA, technical, fixed two-window break, fit-only two-state diagonal-Gaussian HMM, raw-15D, dictionary, and family-ablation variants; uses a documented stationary bootstrap (Politis & Romano, 1994); and emits Benjamini–Hochberg and Holm adjustments rather than conflating exploratory FDR with confirmatory family-wise control. Its receipted dry run uses eight retained daily datasets (18,160 bars), 544 expanding purged origins, 11 model variants, and three future-only outcomes at 1, 5, and 20 bars. Exactly 53,361 of 53,856 unique cases are scored and 495 not-yet-observable targets remain explicitly pending; all split chronologies pass and no scored row is nonfinite or duplicated. Every HMM fit reached the protocol-specified cap of eight EM iterations without satisfying the source-fixed 10^{-6} relative stopping tolerance, so it is a cap-truncated fit-only comparator rather than a converged latent-state result. Five seeded IID, AR(1), alternating-return, volatility-shift, and missing-volume processes are construction checks only. The retained-cache run is development-only because those data have already informed construction.

A preregistered confirmatory study must still define a frozen universe, start date, option entry rule, holding horizon, and no-touch final holdout. At each origin, all scaling, thresholds, codebooks, and optionality baselines must be refit using only admissible history. The purge/embargo must cover the maximum holding horizon (López de Prado, 2018). Primary endpoints must be stated before inspection and include both statistical and economic measures. The frozen development runner provides a mechanism for this study; it does not make the retained cache untouched.

The current runner implements price/volume technical features, a single-horizon EMA system, a deterministic cap-truncated fit-only two-state HMM, and fixed causal location/scale/range/volume break scores. The last is a break-feature ridge, not a breakpoint estimator. A better-converged or sticky HMM, formal structural-break or Bayesian change-point model, wavelet/scattering, and sliding-window persistent-homology comparators (Perea & Harer, 2015; Gidea & Katz, 2018) remain future because their fitting, transform, prior, scale-bank, and boundary choices must be frozen rather than selected after inspecting this development sample. IV-minus-realized-volatility and cross-market connectedness also remain future because the retained single-symbol OHLCV files lack point-in-time surfaces and aligned panel vintages. Existing ablations remove each field family and compare dictionary labels with the raw continuous vector.

Option outcomes require executable bid/ask observations, commissions, slippage, stale-quote controls, exercise conventions, and normalization by maturity, moneyness, delta, vega, exposure, and holding period. Dependence-aware bootstrap intervals and a predeclared multiplicity procedure must accompany repeated variants; the study must state whether it targets exploratory FDR or confirmatory family-wise error. A positive retrospective average is insufficient for promotion.

K Implementation Scope and Non-Claims

The current repository does not implement a Voss tactical wave, Takens embedding, optical flow, transfer entropy, Granger-causal propagation, Hurst or multifractal estimation, persistent cross-window Form identity, constituent-weighted basket or cross-sectional field models, nine-timeframe fusion, historical option surfaces, or a validated execution model. Pair v1 is limited to two independently constructed fields under one recipe and does not relax those omissions. The renderer uses Canvas2D color quantization and display smoothing rather than true contour geometry, Gaussian/Sobel shaders, or WebGL. These omissions are not equivalent: rendering and many candidate transforms are ordinary engineering, whereas universal state identity, causal fusion, option profitability, and execution quality require new study designs or evidence that cannot be reconstructed after the fact. The repository's FUTURE_WORK_TRIAGE.md records this boundary item by item.

Accordingly, the paper does not claim universal states, attractors, cycles, arbitrage, predictive return skill, profitable option selection, cross-market causation, or autonomous decision quality. The words causal and prospective refer respectively to non-anticipative prefix computation—not causal inference—and time-stamped future evaluation.

L LLM Usage Statement

OpenAI Codex materially assisted ideation, repository and formula inspection, experiment and figure scripting, literature organization, LaTeX drafting, and consistency review. This use follows ICLR 2026 disclosure policy and must also appear in the submission form; the human author remains responsible for all claims, citations, and artifacts.[1] Numerical results come only from executed scripts. A recorded metadata audit covers all 38 bibliography entries and reconciles three registry-format discrepancies. No language-model output is empirical evidence or authorship.

[1] https://iclr.cc/Conferences/2026/AuthorGuide; https://iclr.cc/FAQ/LLM